10 Biology: Evolution

ACSSU185: The theory of evolution by natural selection explains the diversity of living things and is supported by a range of scientific evidence

ACSHE191: Scientific understanding, including models and theories, is contestable and is refined over time through a process of review by the scientific community.

Lesson	ACSSU/SHE Elaborations	Content	Success Criteria	Activities
1 - 2	• Describing biodiversity as a function of evolution	**Introduction to evolution:** • What do we already know about the theory of evolution? • How does evolution work? **Darwin & Wallace** • Discoveries & contributions • Opposing theories (Lamarck) • Controversy surrounding theories • Galapagos Finches, Tortoises & Iguanas	• Describe the contributions of Charles Darwin and Alfred Wallace to the theory of natural selection • Describe the evolutionary theories and work of other scientists such as Lamarck and Wiseman • Understand that scientific theories are based on well-substantiated evidence that are contested and refined over time.	• **Kurzgesagt**: How evolution works: https://youtu.be/hOfRN0KihOU • Have a discussion on why evolution will only ever be a theory rather than fact. • Stile: Introduction - Evolution
3 - 4	• Outlining processes involved in natural selection including variation, isolation & selection • Investigating changes caused by natural selection in a particular population as a result of a specified selection pressure	**Natural Selection is the mechanism of evolution** • Darwin's observations • Variations in populations • Allele frequencies • Mutating Moths (Peppered Moth) **Genetic Drift** **Gene Flow** **Migration**	• Define the terms natural selection, allele frequencies, gene pool, and mutations • Describe how natural selection results in permanent changes in the frequency of alleles in a population • Understand how mutations introduce new alleles into a gene pool • Discuss how selection pressures change the allele frequency of populations over time.	• Experiment 2.2 • Darwin's finches' worksheets • Stile: Gather - Evolution
5 - 6		**Different selection pressures cause divergence. Similar selection pressures cause convergence.** • Galapagos Finches • Sharks & Dolphins	• Define the terms divergent evolution, convergent evolution, speciation, homologous structure and analogous structure • Explain how different selection pressures can cause divergence and the formation of new species (speciation) • Explain how similar selection pressures can cause convergence • Identify homologous and analogous structures when comparing the evolution of different organisms	• Experiment 2.3 • Analyse the wings of different animals on the National Geographic website
7	**ASSESSMENT 5 - Modelling Natural Selection in Beetle Populations (5%)**			
8 - 9	• Evaluating and interpreting evidence for evolution, including the fossil record, chemical and anatomical similarities, and geographical distribution of species	**Evidence for evolution** • Fossils • Fossil dating (stratigraphy / radio-carbon)	• Discuss why fossils are primarily found in sedimentary rocks. • Explain how the existence of fossils provides evidence to support the process of evolution • Describe how fossils are dated using relative and absolute dating techniques • Explain how transitional fossils and living fossils provide evidence for evolution	• Review previous learning – different types of rocks and processes such as erosion • Experiment 2.4 • Watch 'Great transitions' video on Archaeopteryx • Use an image of an Archaeopteryx fossil for students to identify key features which identify it as a transition fossil.

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10 - 11	• Evaluating and interpreting evidence for evolution, including the fossil record, chemical and anatomical similarities, and geographical distribution of species • Considering the role of different sources of evidence including biochemical, anatomical and fossil evidence for evolution by natural selection	**Other Evidence for evolution** • Biogeography • Vestigial Structures • Embryology • DNA & proteins **Phylogenetic trees**	• Explain how continental drift provides a well-supported explanation for the geographical isolation of species that eventually results divergent evolution • Describe similarities during the early stages of embryo development in different species • Identify vestigial structures and explain how they are interpreted as evidence of an ancestral heritage in which these structures once performed other tasks • Describe the universal nature of DNA and proteins • Understand how mutations can cause small differences that can accumulate over time • Explain how scientists use the differences in DNA sequences to compare evolutionary relationships between species • Interpret and draw basic phylogenetic trees.	• Use a map with continents that students can cut out and demonstrate continental drift. • Have images of different flightless birds which students to cut out and stick on their maps to show their positions prior to separation. • Research vestigial structures in humans and compare the function of structure in other organisms. • Review protein synthesis – students can practise transcribing DNA into RNA and translating RNA codons into amino acid sequences. • Experiment 2.6
12	• Investigating changes caused by natural selection in a particular population as a result of a specified selection pressure such as artificial selection in breeding for desired characteristics	**Humans artificially select traits** • Selective breeding • Artificial Selection • Evolution of super-bacteria	• Describe how the breeding of organisms with desirable traits has led to domestication • Explain how the process of artificial selection occurs in species such as dogs • Explain how the misuse of antibiotics has led to the evolution of super bacteria such as MRSA	• Experiment 2.7 • Compare and contrast the process of natural selection and artificial selection using a Venn diagram or concept map. • Research artificial selection in crops such as corn, link to molecular evidence for evolution.
13		**Natural Selection affects the frequency of alleles** • Sickle Cell anaemia • Selection pressures	• Account for differences in the frequency of the sickle cell allele in different regions around the world • Explain how malaria is a selection pressure for the sickle cell anaemia allele • Outline how the process of natural selection can result in an increase in the frequency of the sickle cell allele in malaria prone regions	• Experiment 2.8 • Create a crossword using all the key terms used based around natural selection. • Watch a short film about increased frequencies of the sickle cell allele in malaria prone regions around the world.
14		**Climate Change & Natural Selection**	• Analyse the potential effects of climate change on the species of earth, providing evidence to support ideas • Suggests ways to assist species in their avoidance of going extinct • Predict the possibility of new species that may evolve as a result of climate change	• Create a mindmap ideas about how climate change could affect species (plant and animal) • Draw/animate a predicted 'new species' or 'evolved species' that could develop as a result of natural selection
15	• All of the above	**Revision** • Key words • Review questions from chapter 2	• Review all previous success criteria	• Students can make dominoes with Key Words on one end and definitions/diagrams/examples on the other end • Students can create mind maps, Venn diagrams or other graphic organisers to summarise the key concepts of this chapter • Peer teaching: students can work in groups to reteach the content of the unit to the class for the purpose of revision. Each group could be allocated a double-page to summarise
16			**ASSESSMENT 6 - Topic Test (5%)**	